Medical Statistics – lecture 6
Correlation and linear regression

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Part I

- Difference between descriptive and inferential statistics
- Principles of statistical inference
- Tests for comparing a continuous outcome between two or more groups
- Tests for comparing a categorical outcome between two or more groups
- Sample size determination

Today

- Conceptual set-up: association between two random variables
  - Independence, conditional distributions, and co-variation
- Measures for describing an association between two continuous variables
  - Pearson's correlation coefficient
  - Spearman's rank correlation coefficient
- Linear regression to model how a continuous outcome variable changes in response to changes one or more explanatory variables
  - Simple linear regression
  - Multiple linear regression

Joint distribution

Definition

The joint distribution of two random variables X and Y describes the probability of each pair of values (x, y)

Notation

X = smoking status
Y = chronic cough status
Cell: P(X=x, Y=y)

Smoking status	No chronic cough	Chronic cough	Marginal
Never smoker	**0.38**	**0.04**	0.42
Former smoker	**0.23**	**0.07**	0.30
Current smoker	**0.15**	**0.13**	0.28
Marginal	0.76	0.24	1.00

Example question

What is the probability that a person is a current smoker and has chronic cough?

P(current smoker and chronic cough) = 0.13

Marginal distributions

Definition

A marginal distribution gives the probabilities for one random variable, X or Y, ignoring the other

Notation

Row margin: P(X=x)
Column margin: P(Y=y)
Sum cells over the other variable

Smoking status	No chronic cough	Chronic cough	Marginal
Never smoker	0.38	0.04	**0.42**
Former smoker	0.23	0.07	**0.30**
Current smoker	0.15	0.13	**0.28**
Marginal	**0.76**	**0.24**	1.00

Example questions

What proportion are current smokers? P(current smoker) = 0.28
What proportion have chronic cough? P(chronic cough) = 0.24

Independence

Definition

Random variables X and Y are independent if knowing the value of one does not change the distribution of the other

Notation

$P(X=x, Y=y) = P(X=x) \times P(Y=y)$
for all values x and y

Smoking status	No chronic cough	Chronic cough	Marginal
Never smoker	0.38	0.04	0.42
Former smoker	0.23	0.07	0.30
Current smoker	0.15	**0.13**	**0.28**
Marginal	0.76	**0.24**	1.00

Example question

Would smoking status and chronic cough be independent?
Here, independence would mean smoking status does not change the probability of chronic cough.
Expected joint: $0.28 \times 0.24 = 0.067$; observed joint = 0.13 → not independent.

Conditional distributions

Definition

A conditional distribution describes the distribution of one random variable within a fixed value of the other

Notation

P(Y=y | X=x) = P(X=x, Y=y) / P(X=x)
If independent: P(Y=y | X=x) = P(Y=y)
If not: Y's distribution depends on X

Smoking status	No chronic cough	Chronic cough	Marginal
Never smoker	0.38	0.04	0.42
Former smoker	0.23	0.07	0.30
Current smoker	**0.15**	**0.13**	**0.28**
Marginal	0.76	0.24	1.00

Example question

Among current smokers, what proportion have chronic cough?
P(Y=chronic cough | X=current smoker) = 0.13 / 0.28 ≈ 0.46
Compare with P(Y=chronic cough) = 0.24: not independent.

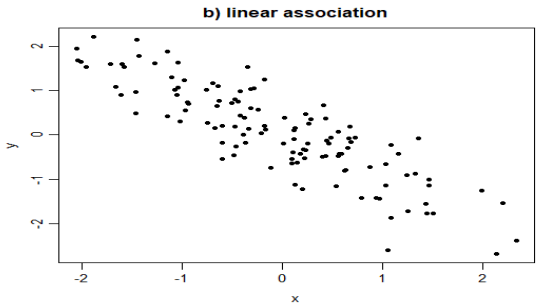
Association can be assessed in different but related ways

- In this course, association means two random variables are not independent: knowing one gives information about the other
- Methods differ in whether they focus on factorisation, conditional distributions, or co-variation

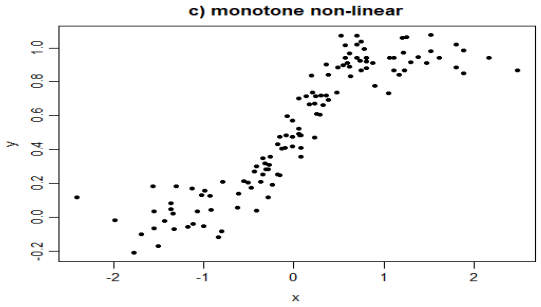
Lens	Core question	Examples of methods
Independence / factorisation	Does the joint distribution factor into the marginals?	Chi-square test of independence
Conditional distributions / group comparisons	Does the distribution of Y change across values of X?	t-test / ANOVA Chi-square homogeneity Linear regression
Co-variation	Do X and Y move together systematically?	Correlation

Patterns of systematic co-variation

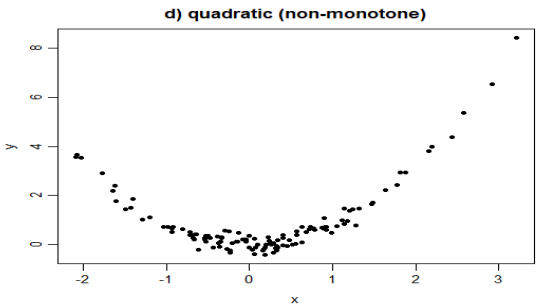
- For two continuous random variables, association often appears as systematic co-variation: values of Y change in a patterned way across values of X
- The form of the pattern matters: it may be linear, monotone but curved, or non-monotone
- Scatterplots are the first diagnostic: they show the form, direction, strength, and possible outliers in the sample data
- Correlation summarises co-variation with a number: Pearson focuses on linear association, while Spearman focuses on monotone association



Linear: X and Y tend to vary together along an approximately straight-line pattern



Monotone: one variable tends to increase or decrease with the other, but not at a constant rate



Non-monotone: there is a systematic relationship, but the direction changes

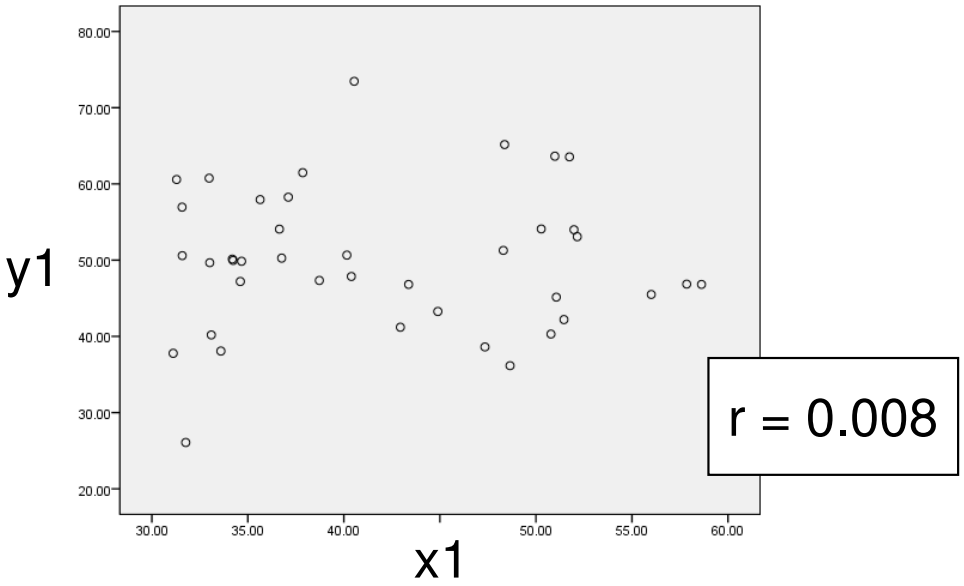
Pearson's correlation coefficient

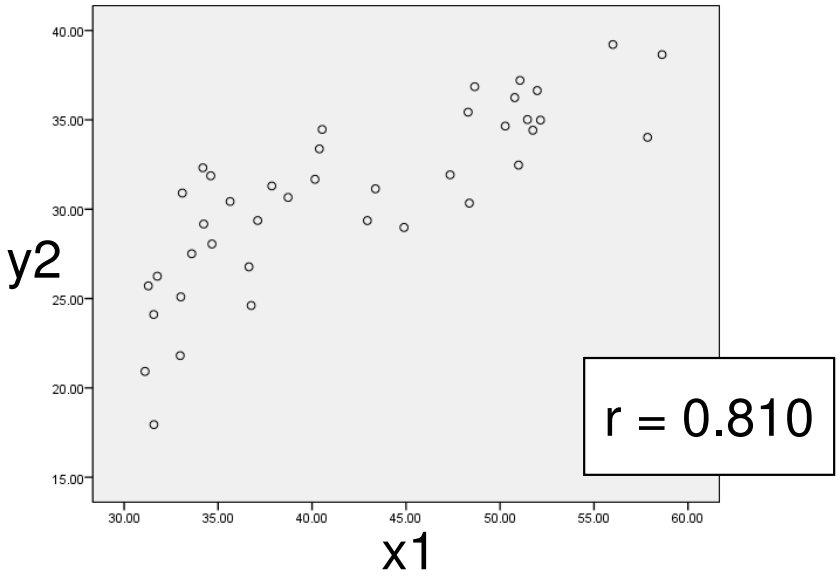
- Pearson's correlation coefficient (r) summarises the extent to which two continuous variables co-vary in a linear pattern
  - Scale: r ranges from –1 to +1, with the sign indicating direction and the maginitude indicating strength

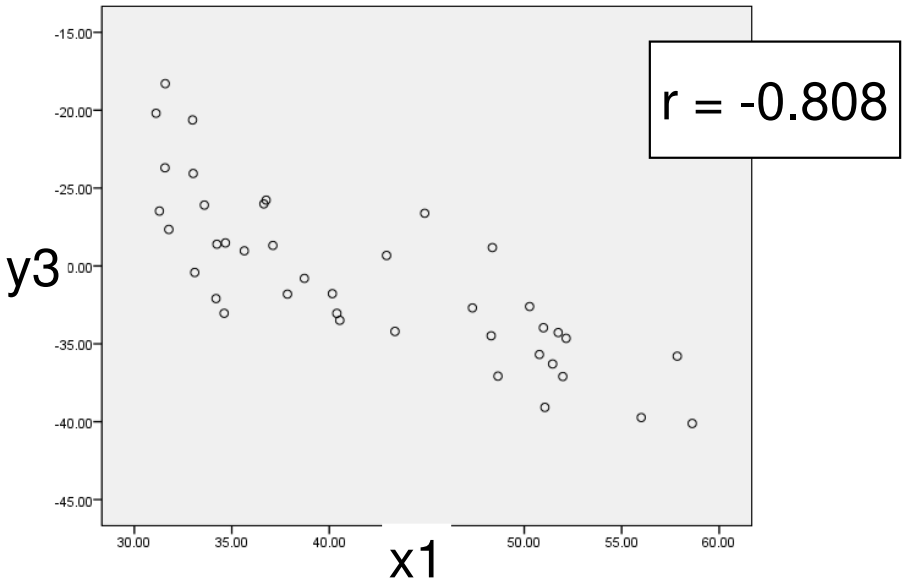
- The sign of r reflects the direction
  - positive: variables tend to vary in the same direction
  - negative: variables tend to vary in opposite directions

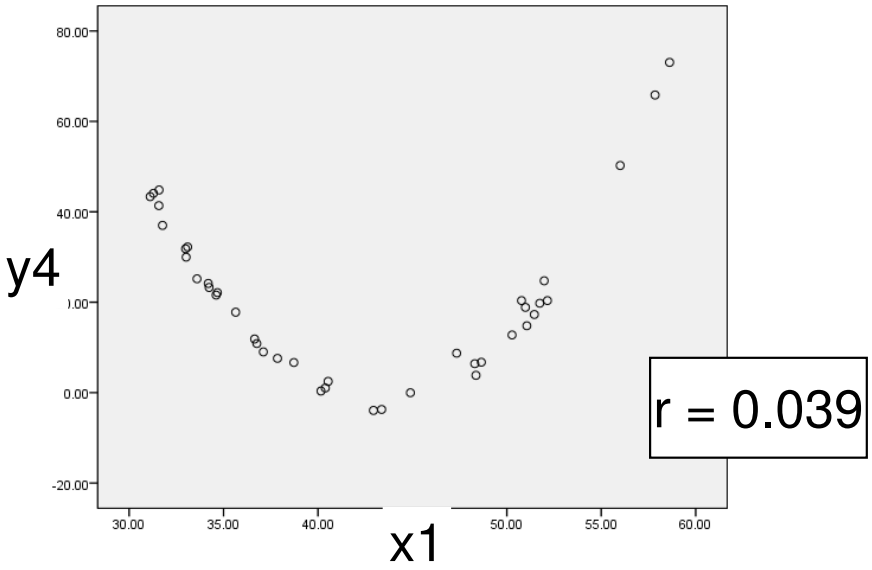
- The magnitude of r reflects the strength
  - values close to ±1 indicate a strong linear association
  - values close to 0 indicate a weak linear association

Some correlation coefficients









Pearson's r is a measure for linear association

Spearman's rank correlation coefficient

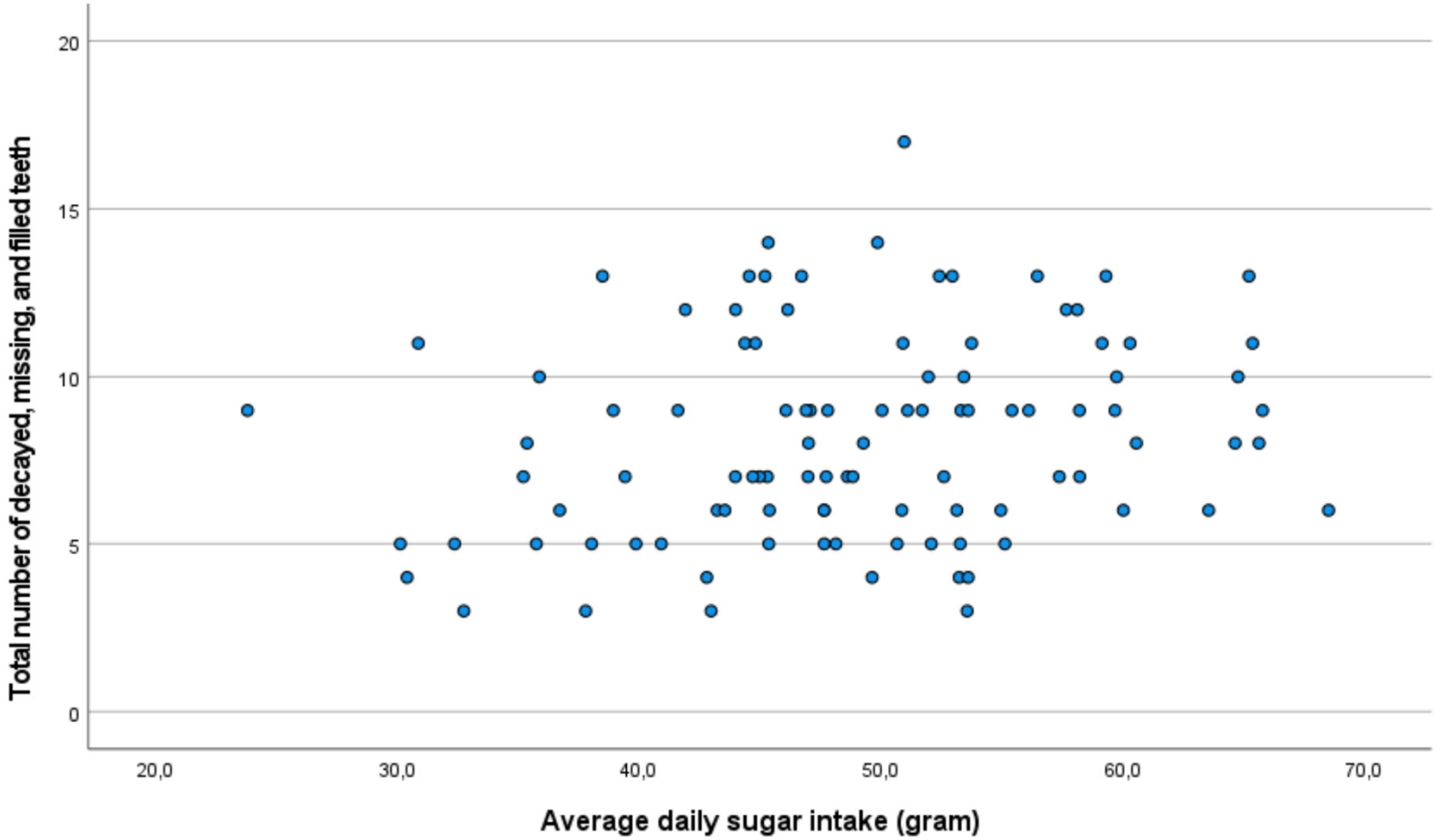
- Spearman's rank correlation coefficient (rho) measures association after replacing the data values by their ranks
- It captures monotone association: as X increases, Y tends to consistently increase or consistently decrease, but not necessarily in a straight-line pattern
- Because it uses ranks rather than the original values, it is less affected by extreme values and skewed measurement scales than Pearson's correlation
- It can be used for ordinal variables, or for continuous variables when the relationship is monotone but non-linear

From correlation to linear regression

- Earlier, we used Pearson's correlation coefficient to summarise the strength and direction of linear co-variation between two continuous variables
- Correlation treats the variables symmetrically: it does not distinguish an outcome from an explanatory variable
- Simple linear regression keeps the idea of a linear relationship, but introduces a direction:
  - One variable is treated as the outcome Y
  - The other is treated as the explanatory variable X
- Regression describes how the conditional distribution of Y changes across values of X

Linear regression - example

How does the expected DMFT score vary depending on an individual's average daily sugar intake?



Simple linear regression model

DMFT-score = $\beta_0 + \beta_1$(daily sugar intake) + ϵ

- **Linear predictor (green):** the expected DMFT score at a given level of sugar consumption
  - Captures the systematic information that the daily sugar intake provides about the DMFT score

- **Assumption:** the expected DMFT score increases **linearly** with the daily sugar consumption (straight line):
  - β_0 is the intercept of the line with the y-axis
  - β_1 is the slope of the line

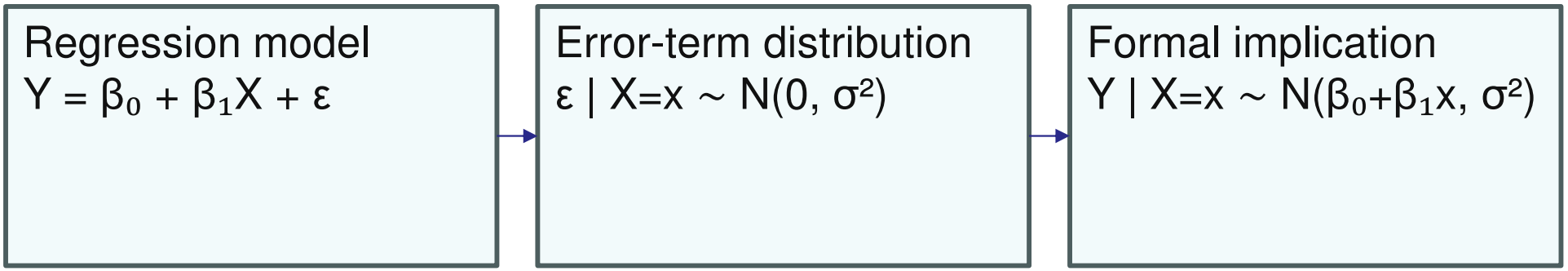
Simple linear regression model

DMFT-score = β_0 + β_1(daily sugar intake) + ϵ

- **Error term (red):** random fluctuations of the observed DMFT score from the expected DMFT score at a given level of daily sugar intake
  - Captures the influence of variables other than daily sugar intake score on the DMFT score

- **Assumption:** the errors are normally distributed with mean 0 and constant variance σ^2

Error-term distribution implies conditional distribution

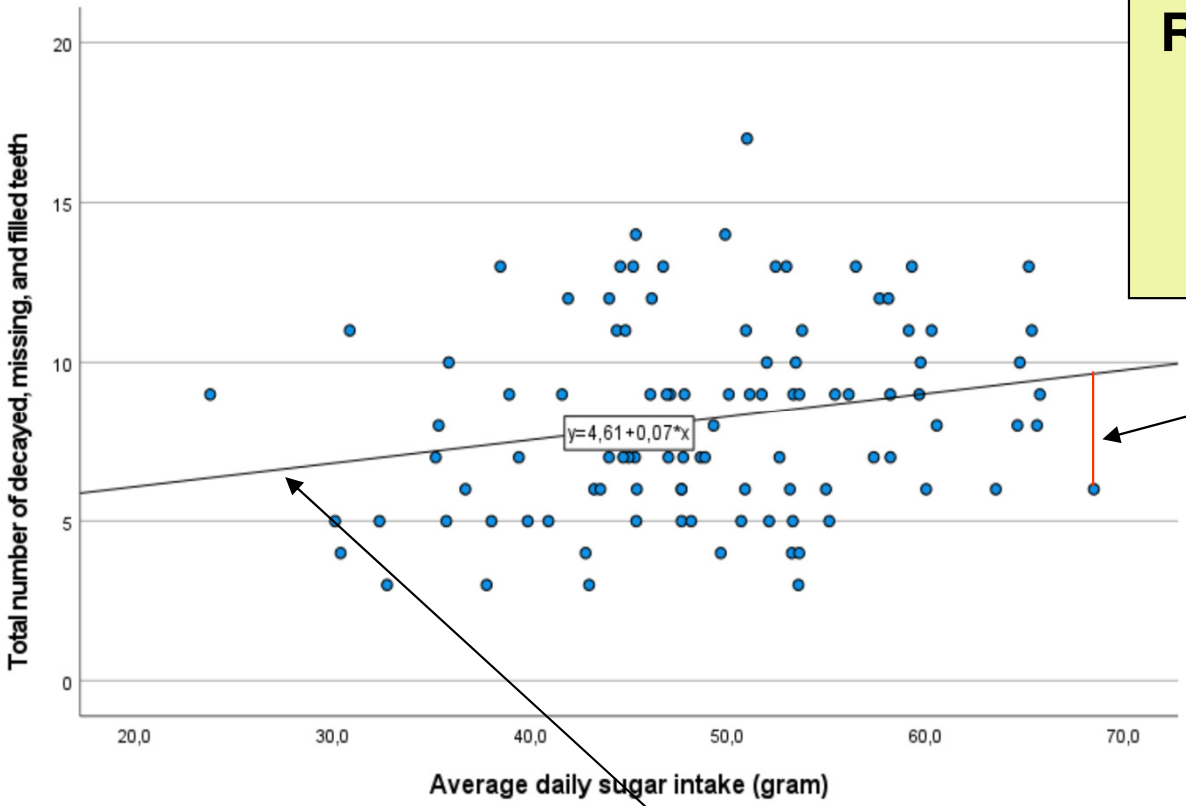


Interpretation

Linear regression is not only a line: it specifies how the outcome is distributed at each value of the explanatory variable

- $\beta_0 + \beta_1 x$ is the conditional mean of Y
- σ^2 is the conditional variance around that mean
- Since $Y = \beta_0 + \beta_1 x + \varepsilon$, a normal error term implies a normal conditional distribution for $Y \mid X = x$

Least squares estimation



Residual: difference between the observed DMFT score and the expected DMFT score based on the fitted model

Best fitting model:
The set of estimated regression coefficients that minimizes the sum of squared residuals

Regression line: graphical representation of the expected DMFT score based on the fitted model

Estimated regression coefficients

Parameter Estimates

Dependent Variable: Total number of decayed, missing, and filled teeth

Parameter	B	Std. Error	t	Sig.	95% Confidence Interval	
					Lower Bound	Upper Bound
Intercept	4,609	1,648	2,797	,006	1,339	7,879
sugar_intake	,074	,033	2,222	,029	,008	,139

Question: Using the fitted model, what is the expected DMFT score for an individual with a daily sugar intake of 50 grams?

Inference for regression coefficients

- Under the linear regression model with normally distributed errors, the least squares estimator of a coefficient has a normal sampling distribution centered at the true coefficient

- The standard error quantifies the sampling variability of the estimated coefficient and depends on the variance of the error term, the variance of the explanatory variable, and the sample size:

$$\mathrm{SE}(\hat{\beta}_1) = \sqrt{\frac{\sigma^2}{(\mathrm{n}-1)\mathrm{Var}(\mathrm{X})}}$$

- The error variance σ^2 is unknown and is replaced by an estimate based on the residuals, resulting in the use of the t-distribution for hypothesis testing and confidence interval calculations
  - Degrees of freedom: sample size – number of estimated regression coefficients (including intercept)

Inference for daily sugar intake

Parameter Estimates

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Intercept	4,609	1,648	2,797	,006	1,339	7,879
sugar_intake	,074	,033	2,222	,029	,008	,139

The estimated regression coefficient for daily sugar intake is 0.074 with corresponding 95% CI ranging from 0.008 to 0.139

This regression coefficient differs significantly from 0 (t = 2.222; p-value = 0.029), meaning that we can conclude that the daily sugar intake is associated with the DMFT score

From statistical significance to explained variability

- Is there evidence of association?
  - The hypothesis test asks whether the observed association is unlikely to be due to random variation alone
  - In the sugar intake example, the p-value suggests evidence of association with DMFT score
- How much variability is explained?
  - A statistically significant model may still explain only a small part of outcome variability
  - R^2 answers how much of the variability in DMFT scores is accounted for by the regression model

R^2: explained variability

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	44.173	1	44.173	4.939	.029[b]
	Residual	876.417	98	8.943		
	Total	920.590	99			

a. Dependent Variable: Total number of decayed, missing, and filled teeth

b. Predictors: (Constant), Average daily sugar intake (gram)

The ANOVA table separates outcome variability into three rows:

- **Regression:** variability explained by the fitted model
- **Residual:** variability left unexplained
- **Total:** total variability in the outcome

R^2 = Regression SS / Total SS = 44.173 / 920.590 = 0.048

Interpretation: the model with average daily sugar intake explains 4.8% of the variation in DMFT scores

In simple linear regression, R^2 equals the square of Pearson's correlation coefficient

F-test for explained variability

ANOVA[a]

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	Total	920.590	99			

a. Dependent Variable: Total number of decayed, missing, and filled teeth

b. Predictors: (Constant), Average daily sugar intake (gram)

On the previous slide, $R^2 = 0.048$: the model explains 4.8% of the observed variation in DMFT scores

The F-test asks whether this explained variability is larger than expected by chance if the true R^2 were 0

ANOVA table: F = 4.939, p = 0.029. In simple linear regression, this is the same evidence as the t-test for the slope

Parameter Estimates

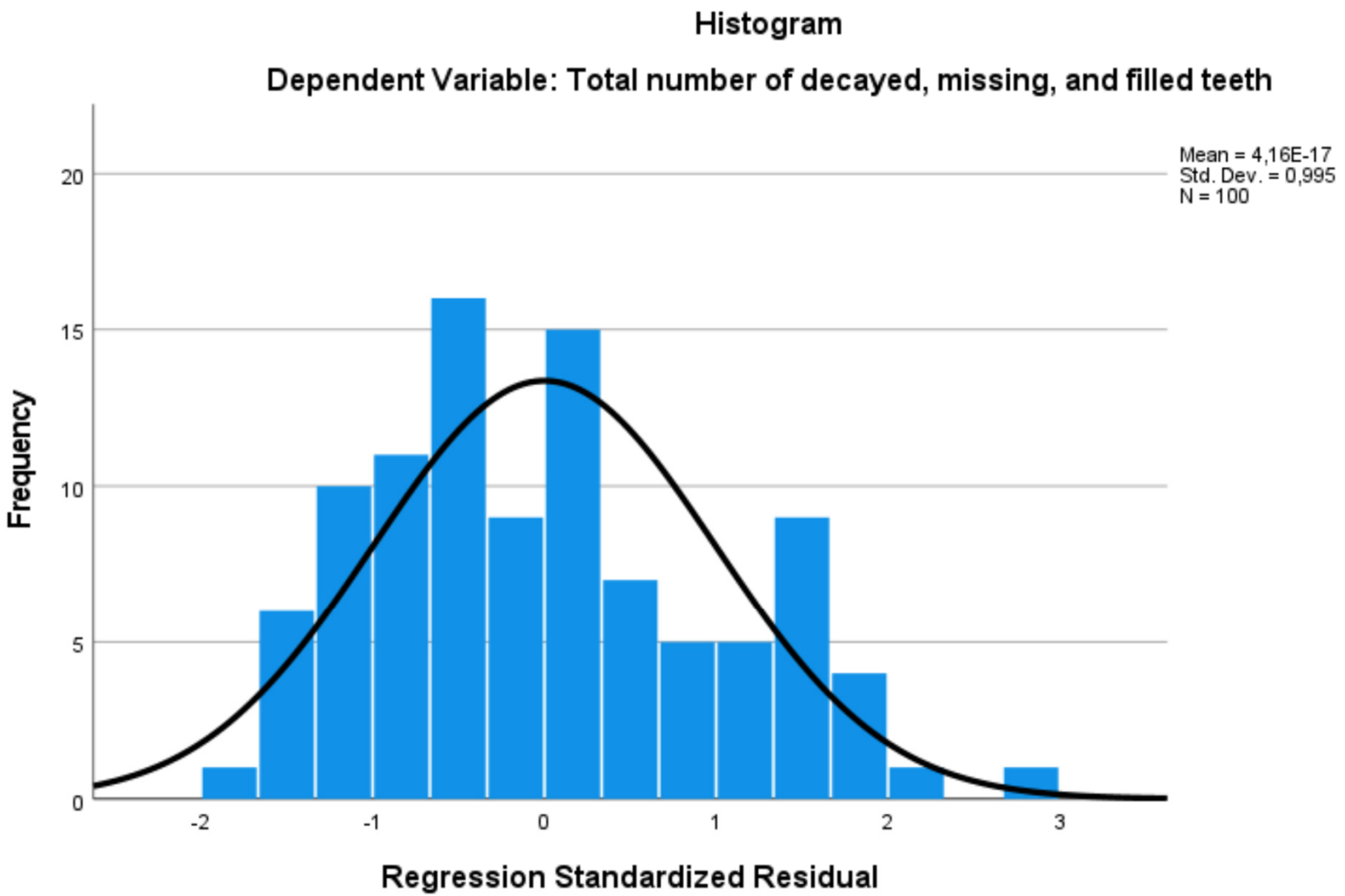
Dependent Variable: Total number of decayed, missing, and filled teeth

Parameter	B	Std. Error	t	Sig.	95% Confidence Interval	
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Intercept	4,609	1,648	2,797	,006	1,339	7,879
sugar_intake	,074	,033	2,222	,029	,008	,139

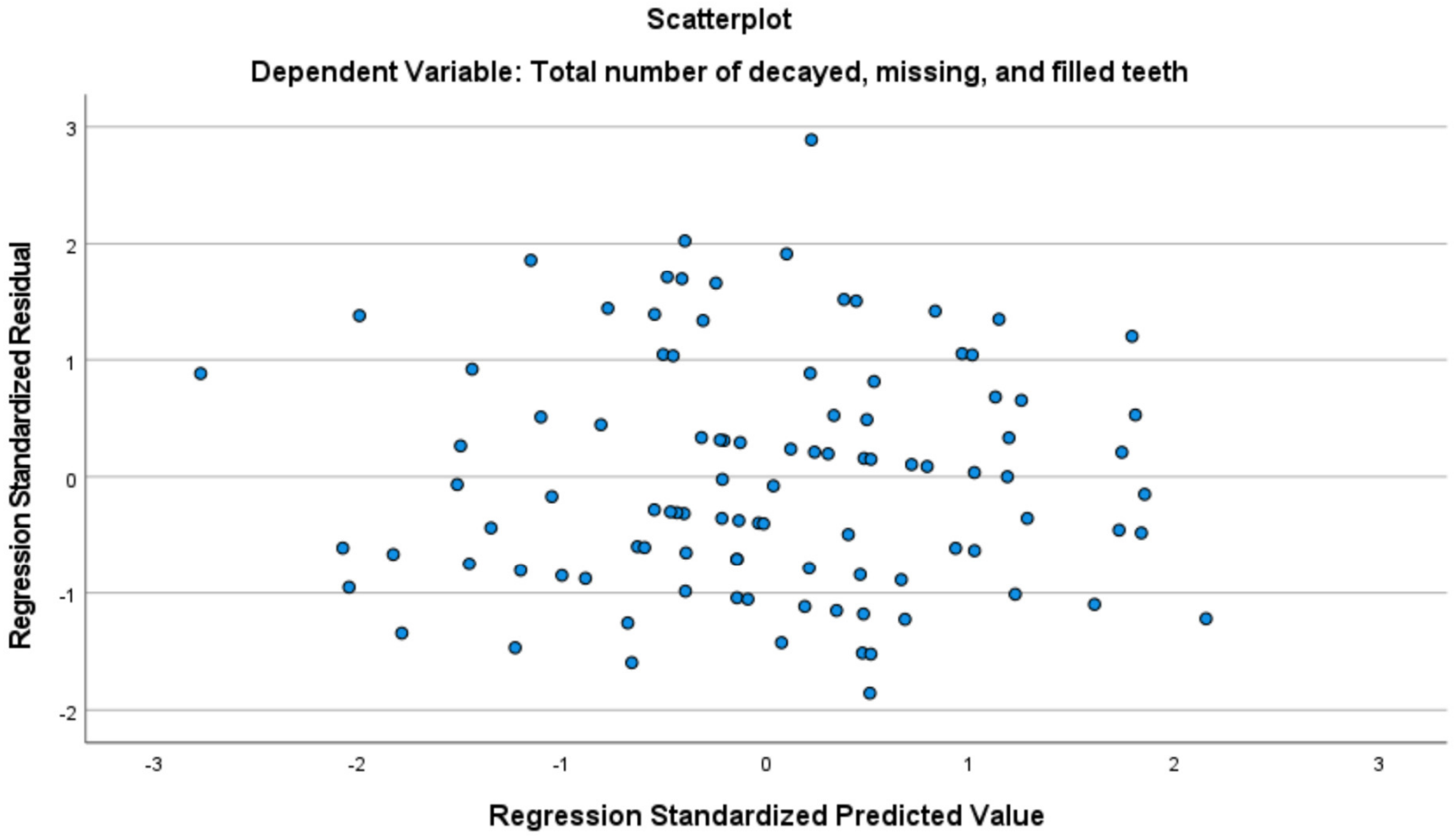
Assumptions in linear regression

- Independent observations
- The relationship between the dependent variable Y and the independent variable X is linear (linear predictor)
  - Note that the independent variable X can be a transformation of another variable X'
  - E.g., the linear predictor $\beta_0 + \beta_1 \log(daily\ sugar\ intake)$ represents a linear relationship between the log of the daily sugar intake and the DMFT score
- Errors are normally distributed around 0
- The variance of the errors is constant (homoscedasticity)

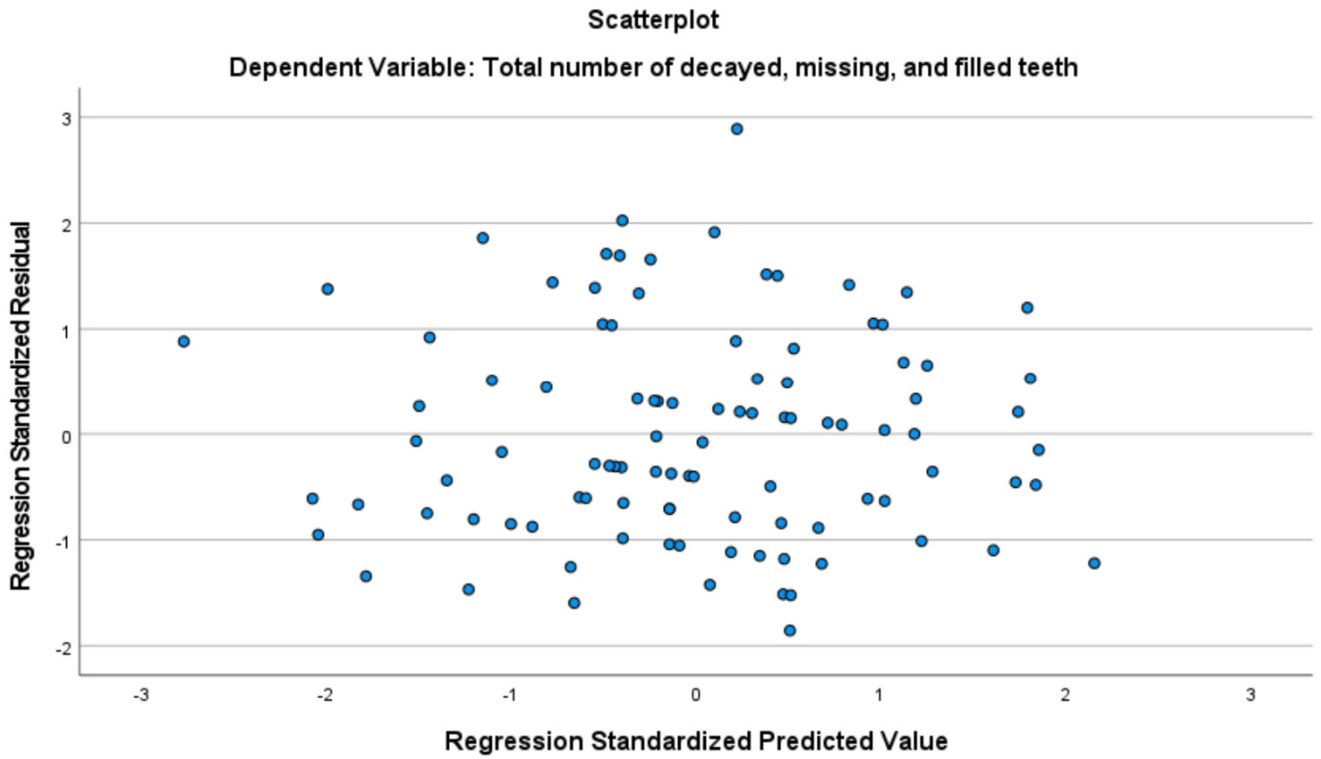
Are the residuals normally distributed?



Is the variance constant?

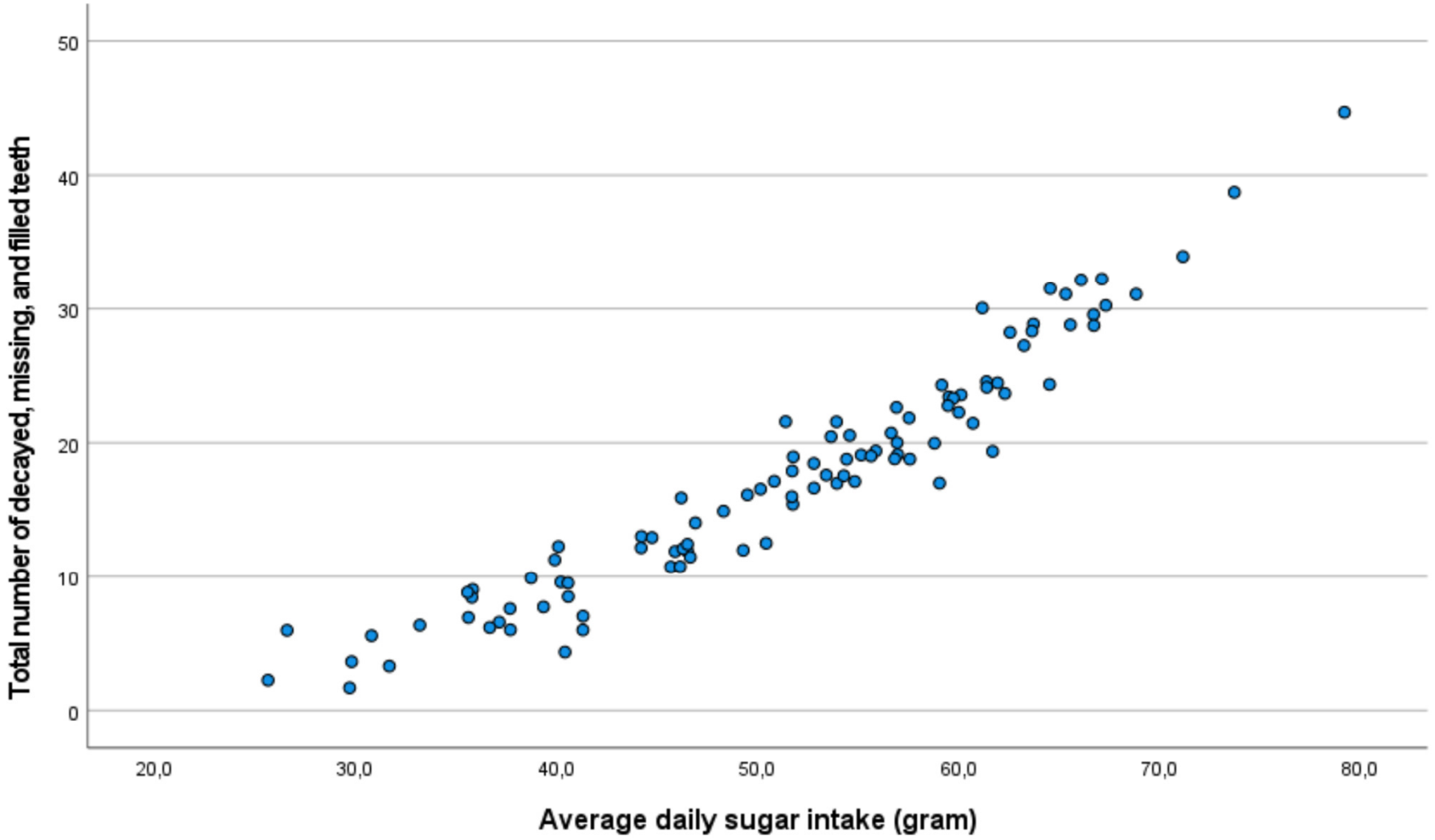


Is the relationship between DMFT score and sugar consumption correctly specified?



- Is there a clear pattern visible in the plot?
- If yes, this indicates model misspecification, and additional or alternative terms should be included in the model

Example of a quadratic relationship

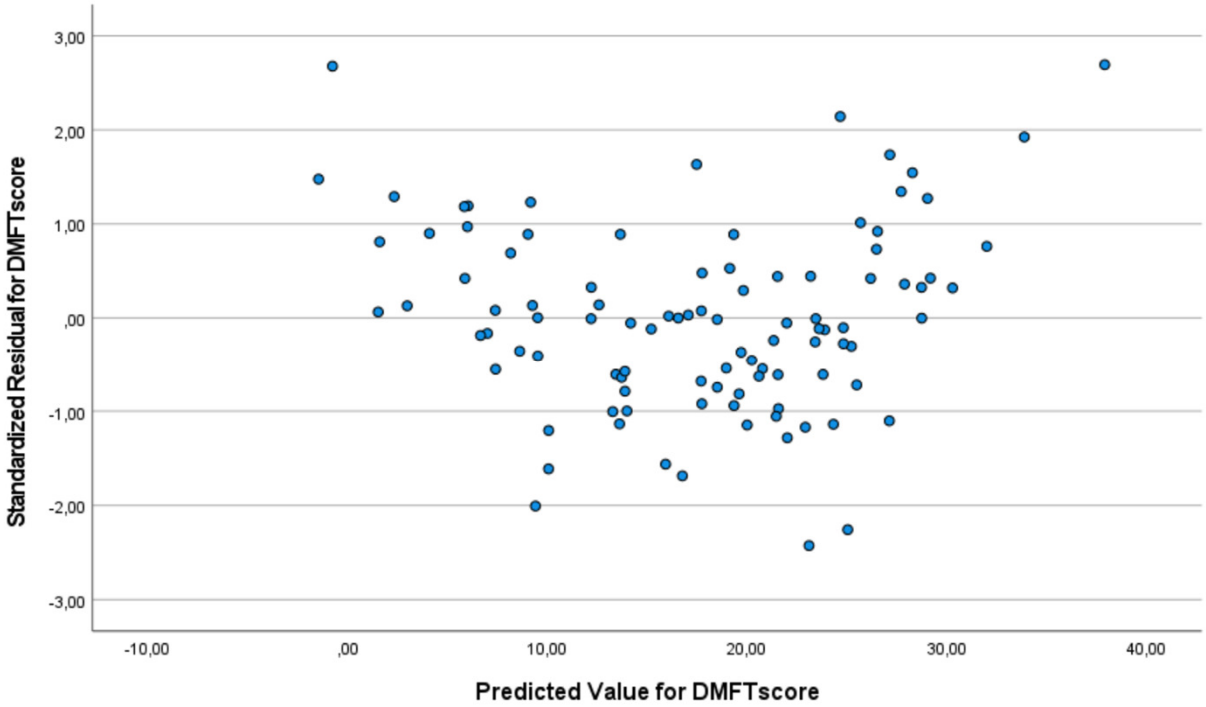


Misspecification: linear relationship

Parameter Estimates

Dependent Variable: Total number of decayed, missing, and filled teeth

Parameter	B	Std. Error	t	Sig.	95% Confidence Interval	
					Lower Bound	Upper Bound
Intercept	-20,303	1,173	-17,309	<,001	-22,631	-17,975
sugar_intake	,735	,022	33,108	<,001	,691	,779



The plot of the standardized residuals against the predicted values shows a quadratic trend

=> Model misspecification!

Fitting a quadratic function

DMFT-score = β_0 + β_1(daily sugar intake) +
β_2(daily sugar intake)^2 + ϵ

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	7153,887	2	3576,944	853,897	<,001[b]
	Residual	406,330	97	4,189		
	Total	7560,217	99			

a. Dependent Variable: Total number of decayed, missing, and filled teeth

b. Predictors: (Constant), sugar_intake_squared, Average daily sugar intake (gram)

Null-hypothesis F-test: $\beta_1 = \beta_2 = 0$

P-value < 0.001 so there is an association between the daily sugar intake and the DMFT score when allowing for a quadratic relationship

Is the quadratic term needed?

DMFT-score = β_0 + β_1(daily sugar intake) +
β_2(daily sugar intake)^2 + ϵ

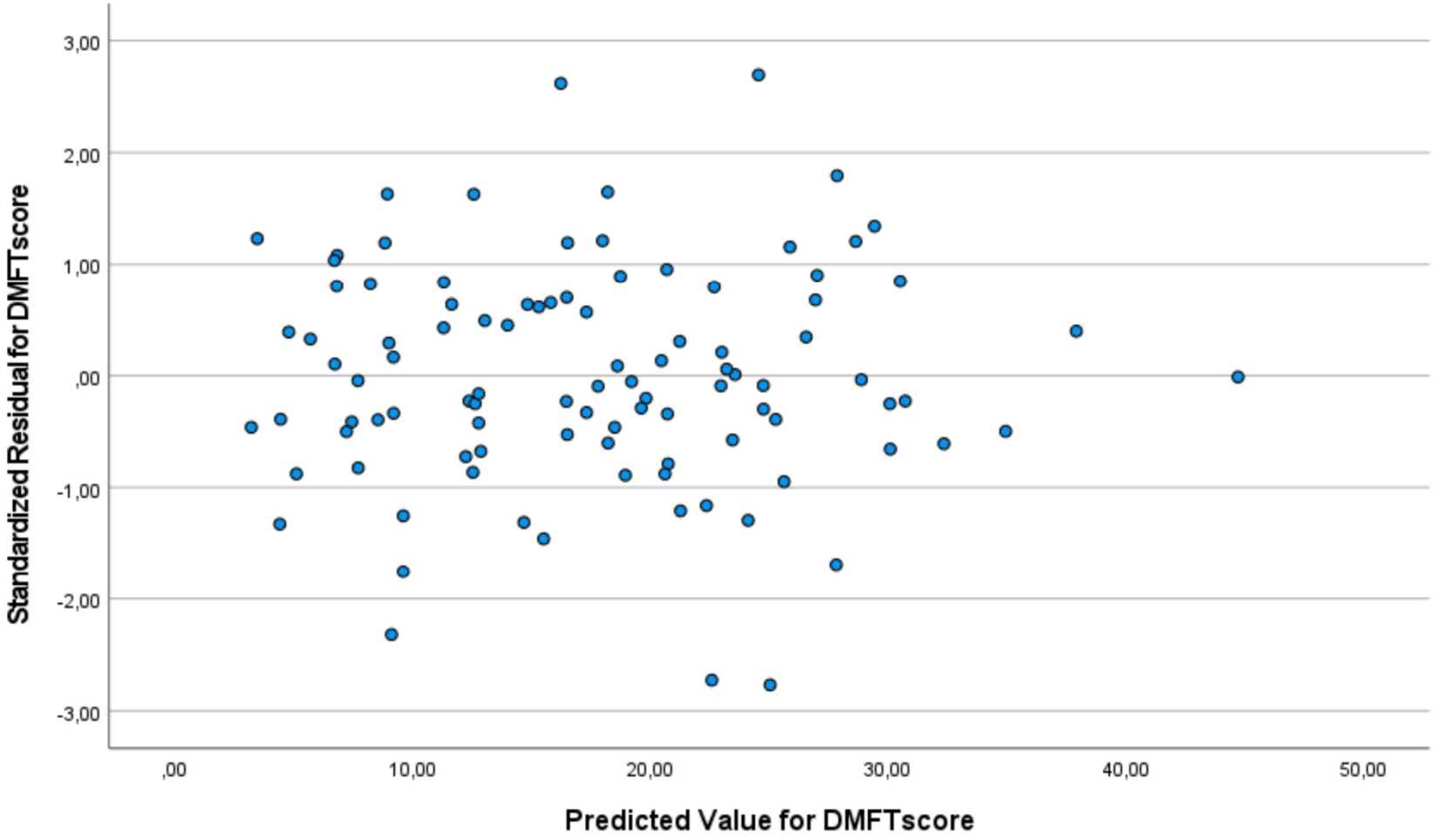
Parameter Estimates

Dependent Variable: Total number of decayed, missing, and filled teeth

Parameter	B	Std. Error	t	Sig.	95% Confidence Interval	
					Lower Bound	Upper Bound
Intercept	3,006	3,397	,885	,378	-3,736	9,748
sugar_intake	-,241	,138	-1,748	,084	-,514	,033
sugar_intake_squared	,010	,001	7,150	<,001	,007	,012

The p-value for the regression coefficient of (daily sugar intake)^2 is < 0.05 so there is evidence that allowing for a quadratic term improves the description of the relationship beyond a purely linear trend

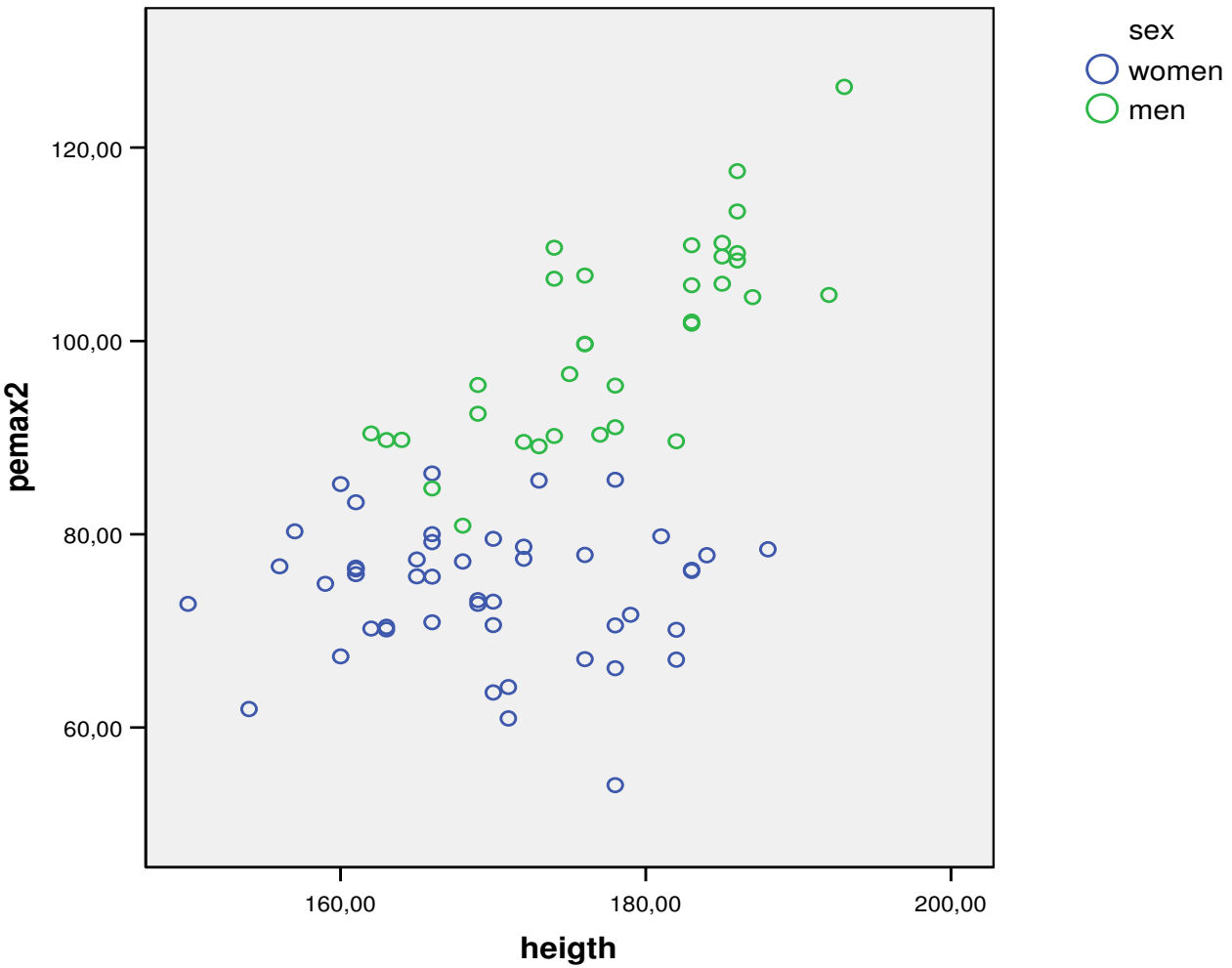
Standardized residuals vs predicted values plot after adding the quadratic term



Some terminology for regression models with a continuous outcome variable

- Simple linear regression: a model with a single numerical explanatory variable
- Multiple linear regression: a model with two or more numerical explanatory variables
- Analysis of variance (ANOVA): a model where all explanatory variables are categorical
  - One-way ANOVA: a single categorical explanatory variable (day 3)
  - Two-way ANOVA: two categorical explanatory variables
- Analysis of covariance (ANCOVA): a model with a numerical and a categorical explanatory variable
  - Can include an interaction term between these two variables
- General linear model: a term encompassing all linear models with a continuous outcome
  - Allows an arbitrary combination of numerical and categorical explanatory variables

ANCOVA – example



Does the maximal expiratory pressure (PEmax, measured in centimeters of water pressure (cm H2O)) depend on the height and sex of the individual?

Coding of categorical variables

- Categorical variables such as sex need to be converted into a numeric format before they can be included in a regression model

- The way in which a categorical variable is converted into numeric values is called the coding scheme

- A commonly used coding scheme is **dummy coding**: each category of a categorical variable is compared to a specific reference group, highlighting differences relative to the reference group

Model specification

Explanatory variables:

Height (cm)

D_sex_male (0 = female, 1 = male)

Regression equation:

$$\text{PEmax} = \beta_0 + \beta_1 \text{Height} + \beta_2 \text{D_sex_male} + \epsilon$$

The dummy variable D_sex_male encodes sex as 0 (female) and 1 (male).

Its regression coefficient (β_2) quantifies the expected difference in PEmax between males and females, with females as the reference group.

Expected PEmax males vs females

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	17,088	17,772		,962	,339
	D_sex_male	23,189	2,018	,745	11,494	,000
	heigth	,337	,105	,208	3,214	,002

a. Dependent Variable: pemax2

Expected PEmax = 17.09 + 0.34*height + 23.19*D_sex_male

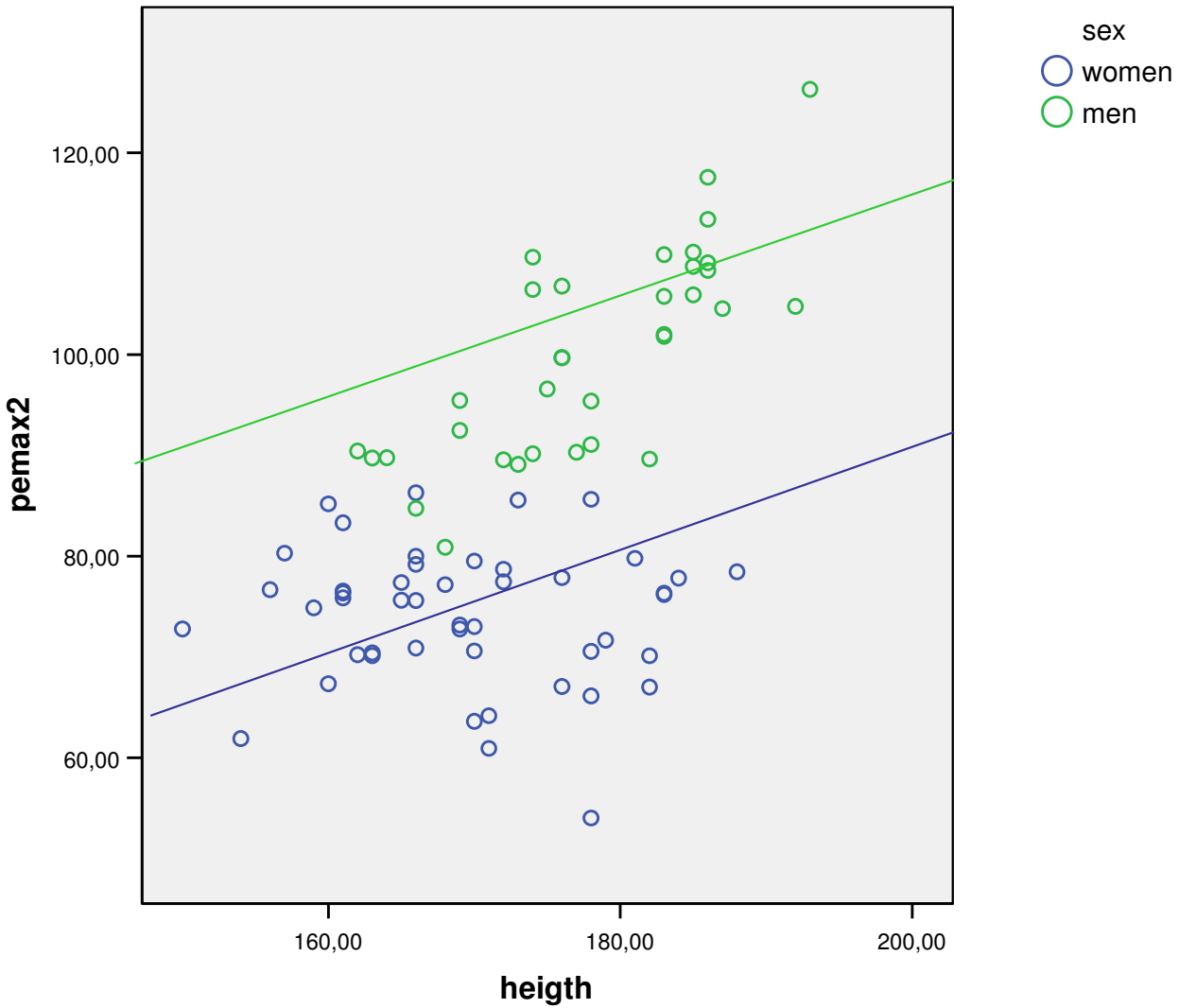
For females, D_sex_male was coded 0, so

Expected PEmax = 17.09 + 0.34*height

For males, D_sex_male was coded 1, so

Expected PEmax = 17.09 + 23.19 + 0.34*height = 40.28 + 0.34* height

Graphically: parallel lines



Interaction (effect modification)

- Previous model: the effect of height on PEmax is the same for males and females

- To allow the effect of height to vary between males and females we add the interaction term ‘height*D_sex_male ’ to the regression model:

  height*D_sex_male = height * D_sex_male

Expected PEmax males vs females

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	82,265	19,241		4,276	,000
	D_sex_male	-153,480	32,173	-4,932	-4,770	,000
	heigth	-,048	,113	-,030	-,425	,672
	Height*D_sex_male	1,012	,184	5,794	5,499	,000

a. Dependent Variable: pemax2

Expected PEmax = 82.27 – 153.48*D_sex_male – 0.048*height + 1.012*height*D_sex_male

For females, D_sex_male was coded 0, so

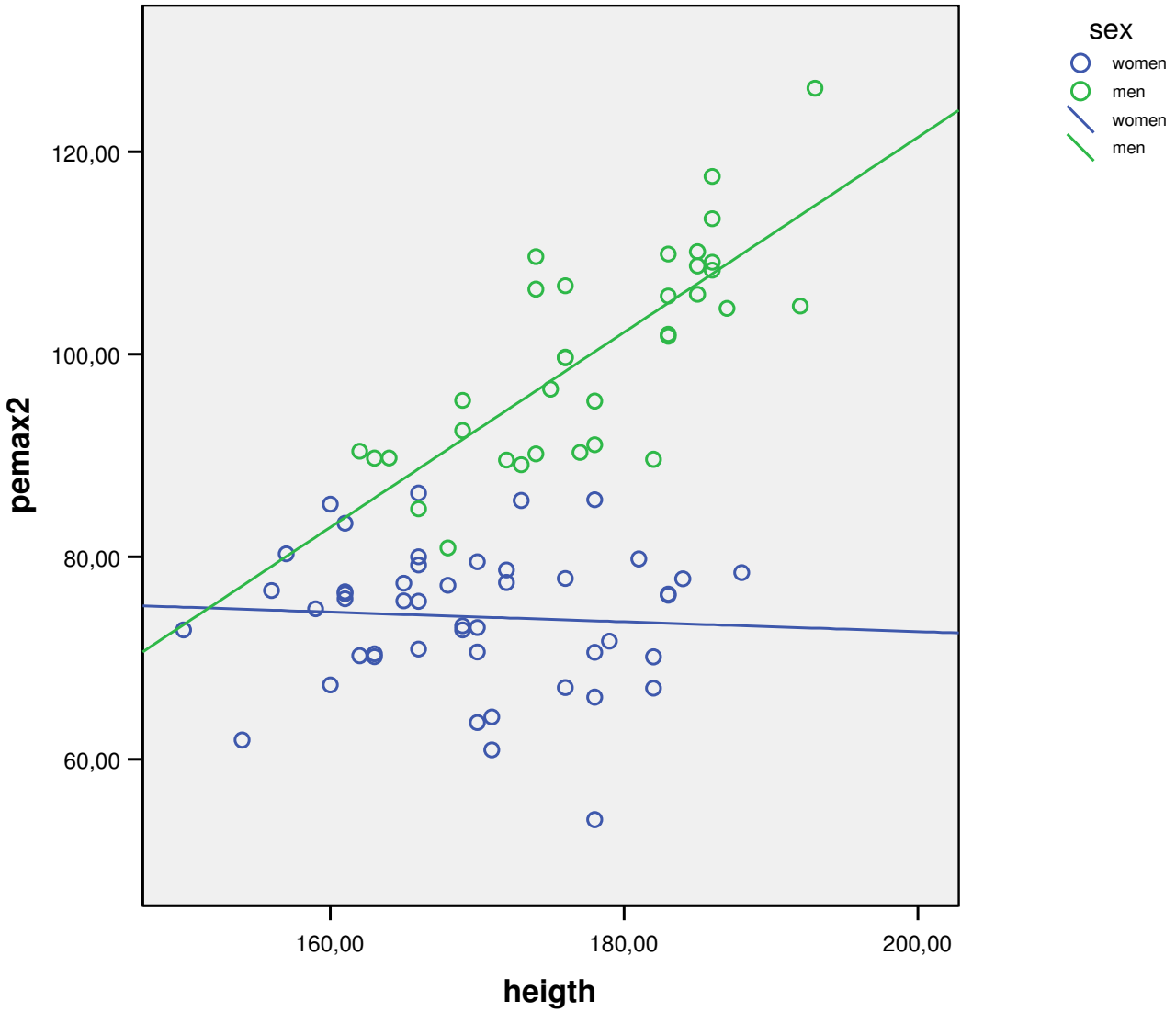
Expected PEmax = 82.27 – 0.048*height

For males, D_sex_male was coded 1, so

Expected PEmax = 82.27 – 153.48 – 0.048*height + 1.012*height =

-71.21 + 0.964*height

Graphically: non-parallel lines



Be wary of the interpretation

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
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	heigth	,337	,105	,208	3,214	,002

a. Dependent Variable: pemax2

Coefficients[a]

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	Height*D_sex_male	1,012	,184	5,794	5,499	,000

a. Dependent Variable: pemax2

Height no longer significant?

Hypothesis testing in linear regression

- For a single regression coefficient, hypothesis testing is based on a t-test
- The t-test assesses whether an individual regression coefficient differs from zero
- In many situations, interest lies in testing hypotheses involving multiple regression coefficients simultaneously
- This leads to F-tests based on the decomposition of sums of squares and the ANOVA table

Comparing nested models

- We have two nested models:
  - Model 1: $X_1, \ldots X_k$
  - Model 2: $X_1, \ldots X_k, X_{k+1}, \ldots X_{k+l}$

- Which model is preferred?
- H_0: $\beta_{k+1} = \ldots = \beta_{k+l} = 0$
- F-test: $F = \frac{(SS_{\text{regression,model 2}} - SS_{\text{regression,model 1}})/l}{MS_{\text{error, model 2}}}$

Covered in more depth during the lab

Questions?